



InGaAs Avalanche Photodiode IFDL-200

IFDL-series

Description

The IFDL-200 is a InGaAs-based avalanche photodiode with a peak sensitivity of 1550 nm and a photosensitive area diameter of 200 μm .



Features

- High speed
- Multiplication gain >30
- 200 μm diameter active area
- Wide operating temperature range

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Applications

- Detection of low optical signals
- LIDAR and rangefinders
- Spectroscopy
- Optical communication system

Generic Characteristics at 25 °C

	Min	Typ	Max
Wavelength range, nm	900	-	1700
Peak sensitivity, nm	1500	1550	1600
Diameter of sense area, μm	-	200	-

Electrical Characteristics at 25 °C

	Min	Typ	Max
Breakdown voltage ($I_d=10\text{ }\mu\text{A}$), V	35	42	55
Responsivity (1550 nm), A/W	-	0,9	-
Gain ($I_d = 1\text{ }\mu\text{A}$)	20	30	-
Breakdown voltage temperature coefficient	0.05	0.10	-
Dark current (Gain=10), nA	-	5	50
Capacitance, pF	-	2	3
Bandwidth (1 MHz, Gain=10), GHz	0.6	0.7	-

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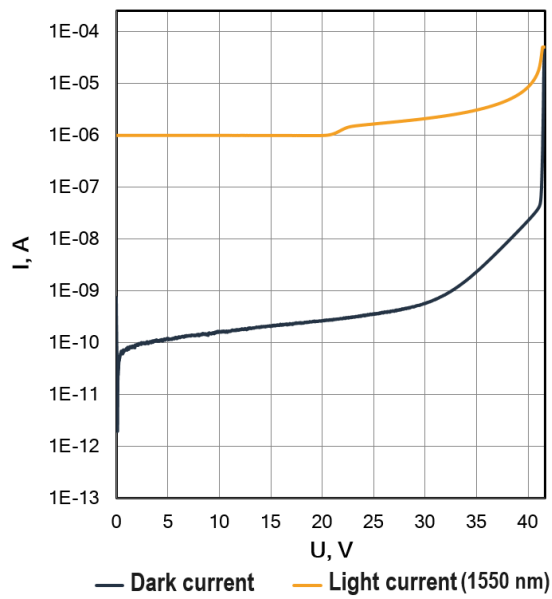


Fig 1. Typical Dark and Light Characteristic

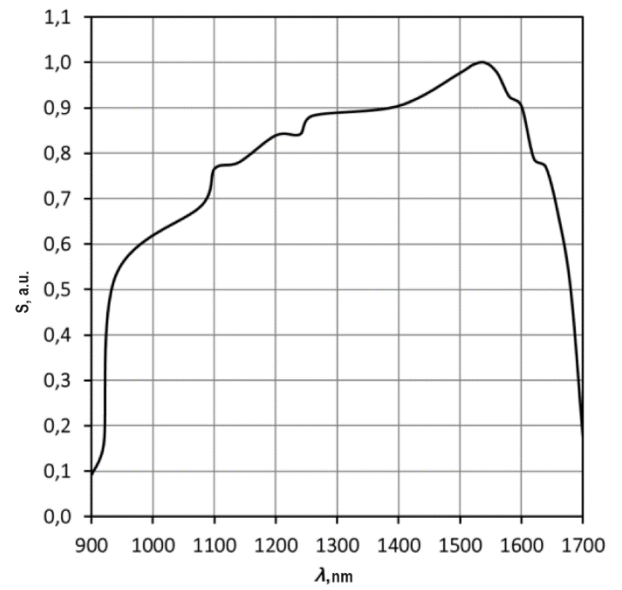


Fig 2. Spectral Response

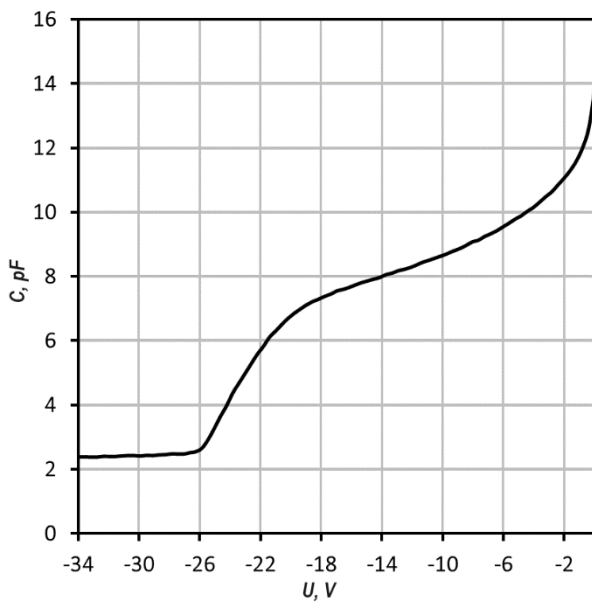


Fig 3. Capacitance vs Reverse Voltage (1 MHz)

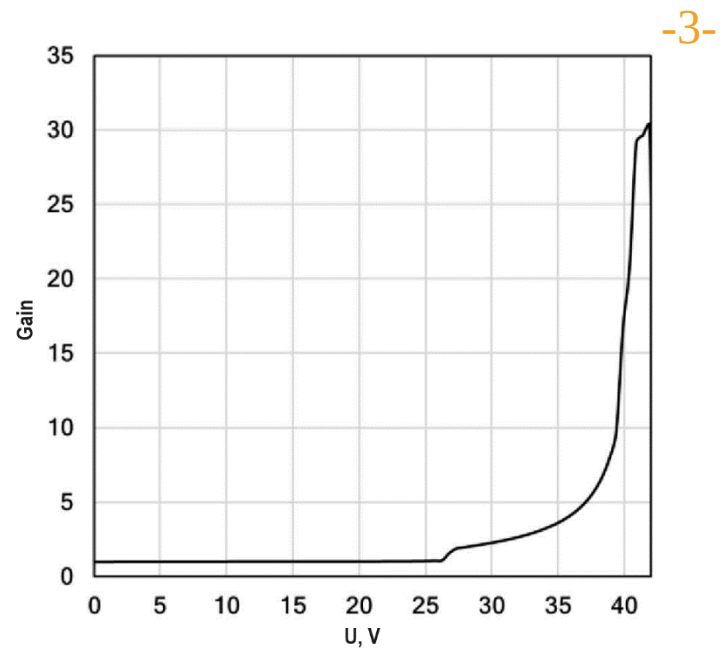
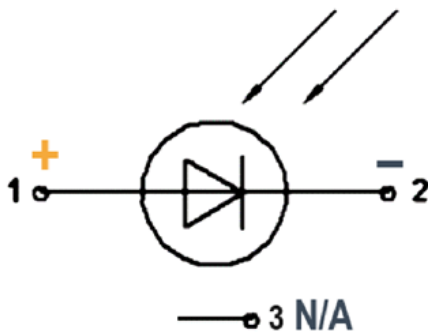


Fig 4. Gain vs Reverse Voltage (1 μW, 1550 nm)

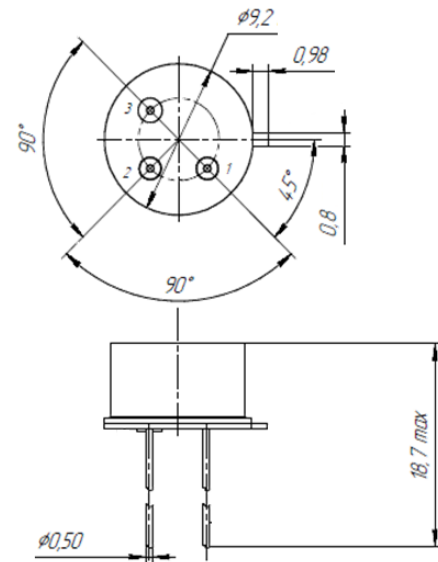


Package drawings

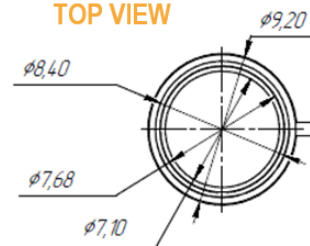
IFDL01A (TO-5 , 3-pin)



BOTTOM VIEW



TOP VIEW



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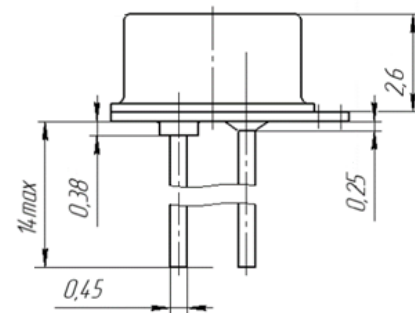
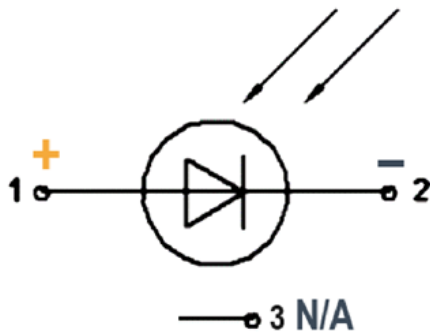
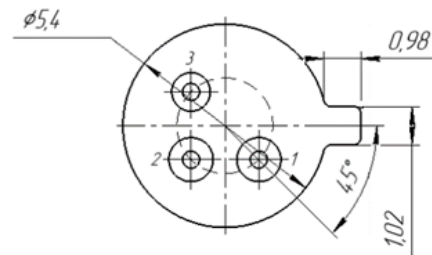


Package drawings

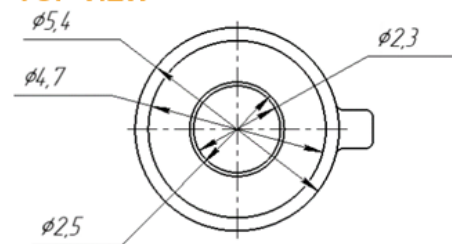
IFDL01A1 (TO-46 , 3-pin)



BOTTOM VIEW

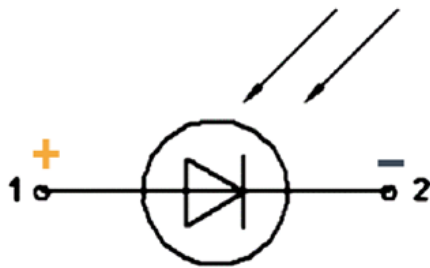
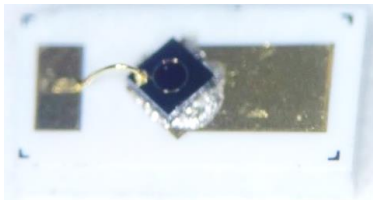


TOP VIEW

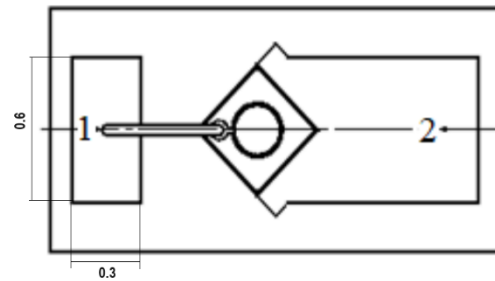




IFDL02-2 (chip on ceramic sub)



TOP VIEW



BOTTOM VIEW

